



UNIVERSITY OF
WATERLOO

MTE 203 – Advanced Calculus

Midterm Exam Fall 2025 – Problem 3

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A function $f(x, t)$ satisfies the following partial differential equation,

$$\frac{\partial f}{\partial t} = \frac{\partial^2 f}{\partial x^2} + e^{-t} \cos x$$

The independent variable x is to be replaced by a new independent variable u , where,

$$u = (x + t^2)^2$$

Derive the differential equation corresponding to the one above which will satisfy the function $f(u, t)$.

Hint: Your final equation should contain only the variables u and t and partial derivatives of f with respect to u and t . That is, all terms in x should be eliminated.

Index: Chain Rule



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